

ActInf Livestream #014.2 ~ The Math is not the Territory: Navigating the Fr

Livestream | The Math is not the Territory: Navigating the Free Energy Principle | 2:11:06

hello and welcome

to the active inference lab this is the

active inference

live stream today we are in active

inference live stream

14.2 on january 26

2021 today is going to be a great

conversation so thanks everyone who's

here live and thanks everybody who's

watching live and in replay

welcome to the active inference lab the

active inference lab as i share my

screen is an experiment in

online team communication learning and

practice related to active inference

you can find us on our website

at twitter gmail youtube or our public

key based user name and uh or public key

based team and shared username

this is a recorded and an archived live

stream so please provide us with

feedback

so that we can improve on our work

all backgrounds and perspectives are

welcome here

and as far as video etiquette for live

stream goes

check out the checklist and remember to

fill out the survey at the end to give

us feedback and then

during the conversation we'll mute and

we'll raise our hands so we can hear

from everybody

to learn more about active inference

live streams for 2021

you can go to this [rb.gui](#) link where
you'll find a spreadsheet
today we are on january 26 14.2 with our
second
participatory group discussion on the
math is not the territory with author
mel andrews
and next week as we head into february
we'll go to a different paper
so check out this spreadsheet if you
want to read the papers ahead of time
and get familiarized with them today
in active infant stream 14.2 we're going
to
go through our introductions and
warm-ups and then we're going to walk
through
14.2 again being a follow-up on the math
is not the territory
navigating the free energy principle by
mel andrews in october 2020
and we're able just to kind of chill
talk raise some questions that come to
mind as we continue
and then next week we're going to be on
a different wavelength
always welcome to have participation so
introductions and warm-ups
it's a nice group of seven today we can
introduce ourself
and feel free to give any kind of
introduction or
check-in you'd like and then we'll pass
it to somebody who hasn't spoken so
i'm daniel i'm in california and i will
pass it to
yvonne hello my name is
ivan i'm from moscow russia and i pass

it to marco

let's go to alex first

[Music]

hi everyone i'm alex i'm also

in moscow russia and i'm a researcher in

systems management school

and i pass it to stephen

hello i'm steven i'm based in toronto

canada i'm doing a practice based phd

at canterbury christ church university

and

i'm going to pass it over to cherry

uh sarah if you want to otherwise let's

go to marco

hi i'm marco uh based in holland and i'm

i'm just a active enthusiast i'll pass

it to

mel hi there

i'm mel andrews i'm doing a phd in

adjacent stuff at the university of

cincinnati

is that it is not all of us yes cool and

then sarah any time you want to

jump in is chill so today oh yeah go

ahead sarah

uh hi i'm sarah i'm a person interested

in stuff and things

mel's talk is awesome here i am what

else can be said

so for the for the warm-up

questions we'll just dwell on them

they're just openings and just

opportunities for us to raise the kinds

of things we've been

thinking about today and so what is the

uh what's something you're excited about

today and then also

what did the article or last week's

discussion

make you curious about

so something you're bringing exciting

today or just a question

or even if it's adjacent or it's not a

quote from the paper or anything like

that

like who are we that has brought us

today to this conversation

marco and then i believe it's stephen

but stephen and mel if you could add

your name on jitsi but marco and then

uh stephen yeah um

well last last week we uh ended the

discussion

with uh please three pill now me and uh

alex keeper on the uh

uh the prospect of continuing discussion

about the relation between a

thermodynamic entropy and information

through

so very much excited to explore that

because i think that was very crucial

parts in the

paper that deserve more on

cool steven and then anyone else

yeah i'm i'm bringing it an interesting

um dimensionality and and how that

relates in

like high dimensional spaces low

dimension

spaces and entropy so that that was

something that

this paper really kind of got me

thinking about

so i think while it seems very abstract

okay could you also speak a little bit

more directly into your

microphone um but thanks for that so yes
yes okay so definitely um yes
philosophy paper abstract general
theoretical a lot of adjectives that we
hear with philosophy but then this is
almost about the
philosophy of utility and about the
reality of the models that we
make so it's kind of uh has a hand in
both of those
worlds which is what makes it an
interesting thing to think about i'm
also
curious about that intersection of how
do we generalize and abstract correctly
in order that we can make more effective
action
and i'll just put up the last two
questions and anyone else is
welcome to raise their hand what is
something that you liked or remembered
about the paper
and then what is something that you're
wondering about or would like to have
resolved by the end of the
um paper or by the end of the discussion
today
so one thing i'll just start with is uh
if the math and yvonne you can you too
please thank you so if the math is not
the territory
then what what is our metaphor
here like what is the territory or what
is the math
if we're gonna go with the title of the
paper that
the math isn't the territory what is our
uh

complete spatial metaphor if that's even
helpful to
walk down but it's just um
uh yeah just a few thoughts anyone else
want to raise anything
i i guess i have a maybe a general
question which is um
for mel like uh almost like a
one to ten or how how mel maybe feels
that fep
contributes to like epistemic unfolding
like
the best possible tools to allow us to
ask the most relevant questions
that's
yeah mel yeah i think i so i i think i
i am came at this
with maybe a kind of naive and outdated
view
of of the scientific method
and i really didn't like very
general very abstract models and i
really didn't like
use of metaphor um
because i saw a lot of the the kind of
mistakes that were being made
with very abstract models and with with
kind of analogy
okay please do not yeah thanks
make sure that everyone's audio let's
take a few minutes before
the hour begins so that we're here on
time so that everything is prepared but
thanks a lot the audience also
appreciates it now continue
what happened oh there's just audio on a
loop
cool not me knowers yes not you

although i was also late
yeah so i i i tended to think of um
certain uh
approaches in sciences as being kind of
bad because i saw a lot of
of mistakes that were being made with
them so a lot of um kind of reification
where we start with
an analogy like oh you know we've got
selfish genes
or oh we've got you know borrowing
models from physics into biology you
know and then
and then um confusing them refining
them um but the
the mistakes are not are not the methods
right there's a distinction between the
mistakes that come along with with
certain methods and methods themselves
and there are mistakes that come along
with everything and um
without with any kind of approach you're
going to take complaints
and um especially when it comes
to sort of uncharted waters
um you really need
to be more creative um
you really need more generic models you
need to be borrowing models
from one domain and importing it into
the new domain
kind of casting in that right so when
when you've got um
you know very simple systems and in like
classical physics and chemistry
um you can kind of build up direct
models
um without doing much analogizing right

um but when it comes to dealing with
something like
the human brain or an ecosystem things
you know or even like social structures
were um

[Music]

where we're just sort of out of our gap
you need a lot more
looseness and

for that reason i think when it comes to
the kind of systems that we want to
understand in

future science things like the fcp are
actually going to be

the way to go cool well this whole
future

science i'm sure that's something we can
return to but this was a really

great point that when we're exploring or
constructing a new area

we do need to borrow ideas vocabulary
methods etc because otherwise what is
there to be

discussing or converging on but these
come with baggage preconception

and uh just incompleteness because
they're not the godelian incompleteness
just

they're metaphors they're not total
one-to-one of everything so how do we
navigate borrowing from the past but
also moving into something that has
different attributes

so stephen then anyone else
that's a very good point in terms of
um you mentioned about this like does it
belong to science

so we often think about is something in

engineering or applied science or is it science but then i suppose there's a question of when is something useful to have in a philosophical sense somewhere in between science and philosophy so not completely philosophy but maybe there is another category of um foundational knowing um that we kind of then you then use to build other things upon so maybe there's something in that yeah it's it's almost like this question about whether the natural kinds are disjoint or not and then okay all the stuff or all the approaches that we can take there's science and not science is that a clean break or is there multiple kinds of non-science even multiple kinds of science so this is the demarcation problem and maybe the simplest demarcation would just be uh the bright line and line in the sand but what if the demarcation problem is a little bit more nuanced and it has islands or it has other attributes that are not just a line across the gymnasium floor so mel i'd be curious how did you think about the demarcation problem early on in your research or when you initiated this and then where has that taken you on the demarcation problem

which is again about what is science and
what isn't science
yeah i'm um unfortunately i think the
longer i spend
um my background isn't cognitive science
i was doing
i i had some bio background um and some
psychic concerts um so so this is the
first time i've been
you know full-time a philosopher
um and i find that the longer that i
immerse myself in philosophy
um the more equiney and i become which
is
it sort of to my chagrin i i didn't um
expect to see myself going that
direction what is that
sorry what does that mean the quiney in
in this context
yeah um naturalism
which means no backdoor to knowledge
the um the only access we have to
knowledge
is empirical so
uh there's
there's like science and there's
commentary on science um
and and there's this i view the
relationship as
as between science and philosophy
um as being
kind of intrinsically connected they
need they need each other you know the
philosophy
comes up with new questions and that
they don't know how to ask yet
empirically
and then we figure out over time ways to

go about these

these questions empirically and we we
come up with methodologies and then the
methodologies

fail they hit a they hit a standstill

and then

um philosophers have to come in and say
well look you know you you've kind of
arbitrarily narrowed yourself to the
outset we need to start thinking a
different way right

um and i don't think i think like you
know

in contrast to what what sort of quan
there's a there's a kind of popular
um thought that goes along with
naturalism nowadays which is just that
philosophy is this kind of finite stock
of questions and as we hand off as we
as we pull the questions out of magic
hat and have them off between empirical
sciences

we run out of questions and so
you know fortunately in the near future
philosophers will run out of questions
to ask

because the scientists will have solved
all the questions and then you know then
we'll be out of block right
which i think is just kind of hopelessly
ridiculous

i mean it's it's really a it's a really
simplistic picture and

i'm you know fortunately for me i don't
think

will ever be out of a job i don't think
scientists will ever be out of a job i
don't think philosophers will ever be

out of a job i think i think
there will always be questions for as
long as we persist
um but uh
i i do think that ultimately i am i am
starting to come around
to the idea that there's just an
empirical route to knowledge
and um
that means to me where i'm at with that
right now
in in kind of processing that and
digesting that it's um
that there may not be a sharp dividing
like there might not be a a bright line
distinction
between science and not fights um
and what i really hate about this is is
i
i really want i really want um
math to be entirely conventional i want
math to be
invented not discovered constructed
and to be a naturalist
about mathematics you have to kind of
record it more
reality and naturalness than i i want to
but
you know so that's where i'm at that's
my i might have to but i might have to
bite that bullet
um what what uh stephen was saying
um there's a so the the two people
what you were saying about kind of like
foundational things that come before
science
um this ties into what
uh is being said by uh inez de polito

and um what was said by um matteo
colombo and cory wright
in a 2017 piece um about about first
principles
um so there's a there's a piece 2017
piece called first principles in the
life sciences
something in mechanism um
i think it's in bjps why matteo colombo
and quarter right
about first principles and the fep it's
about the fpp but but
but this first principles approach
in the live chat put it in there so cool
people can check that out
do you want to continue yeah or sarah do
you want to
uh go ahead after you've adjusted your
chair but
go ahead yeah i'm pulling a wire from
under my desk
um yeah i have a question for you mel um
i'm
i'm as a person a new student in
philosophy of science i'm bumping and
somebody who came from physics
i'm i'm bumping up against this like
dynamic that i see between philosophers
and science where
you know science will come up with
something and then there's this kind of
like ankle-biter dynamic where like
philosophers will be like well what
about yeah but what about and i'm just
like
i mean i want a more positive conception
of what philosophy can be and the only
one

so far that i see that's really clear
and i'm a new student so it's not like
so i'm probably missing a lot but like i
really like chris fields because he's
he's more like
not afraid to be sci-fi you know like
he's not afraid to have a speculation
which is
originally what a lot of philosophers
did so i'm like i just i really want
advice as like a student about what
what how you feel that dynamic can be
we'll go mel with a direct ask and then
marco then stephen
yeah chris hughes is awesome he gave a
talk recently to our um
to one of our working groups and that
was fascinating a lot of it was over my
head it's
just introducing so many new concepts
but really fantastic
and he's joining in march early march
for active17
he'll be on yes what are we reading
tbd to be determined but we'll find out
very soon fantastic yes
awesome yeah and there's something i
read
recently i'm there's a long story um
i i wouldn't i don't want anyone
thinking that i would just go out and
read heidegger because i wouldn't
um what happened was there's a long
story about
tracking down quine's typewriter and
and like writing back and forth with
quen's
son and finding a heidegger scholar who

had just acquired the typewriter of
martin heidegger who i used to
correspond with as a kid
um but i ended up reading some height
in in part of as a sort of part of
writing a piece up with a heidegger
scholar
um and uh just just a couple paragraphs
but but there's um
there was this there's a line about um
that uh there's this sort of
um and this is you know this isn't like
at the time of like the vienna
historical at the time of
sort of the the positivism right
and and and there was this line about
logic and philosophy of science kind of
like limping after the sciences and
sweeping things up and putting cobbling
things back together you know like
cleaning up after science
in this very um kind of horrible
lame way um and and it wasn't
contributing
anything of its own it was sort of doing
like you know like logical
reconstruction of
scientific developments but but nothing
that actually really
you know contributed to to furthering
the ends of science and
um haiduker was was declining that
and saying that we need uh a generative
logic
that we need to go back to the model of
um the relationship between philosophy
and science
um of the time of like plato and

uh aristotle and and the
you know thinking ahead of science and i
i i just think
i i very much think that's what we need
and i think to some extent
um there's still very much like a live
tradition in philosophy
that is stuck in this kind of um
we're just we're just describing
we're just doing the bookkeeping right
um
but at the same time there's
i think very much a live and active
tradition of philosophy of science today
that is
that is really thinking ahead of science
and um coming up with new questions
problematizing things
in a way that um that leads to
real breakthroughs cool marco
then steve
thanks yeah i really like the discussion
so far um
i i i i would like to point to the uh
notions of uh abductive inference and
the uh exploration expectation
um um complementarity uh
to maybe approach this relation of
science and philosophy so i mean i
i also have my qualms with philosophy uh
like sarah and
also noted i i personally really hate
the um
excessive well in my in my view a test
of trend to uh
seem authoritative and absolute tears
and these
weird fetishes enough of universal

truths and stuff i i i'm not fun of that
what i'm more fond of
is is the it's almost for me it's an art
right so it's the art of
taking something a given sets of
assertions or propositions whatever
and then generating more territory to
explore from that
right so the way i see it is
for under active inference knowledge is
generative right
and so that generativity can be seen as
leading to abductive inference
so it generates new hypotheses um but it
can also be seen as imagination and
um the that level that that that step
of generativity is then itself
rooted in what's been generated upon a
direct contact with
empirical reality and so for me it's
like an intermediate kind of step it's
like a
a detour you create all these hypotheses
you explore the maps you create it
yourself
and then you you can return to the
imperial grounding
and so that's kind of how i see uh
philosophy it's kind of a deep work
and like mel said they're they're
inextricably connected without an
empirical basis you have no
grounds uh on which to imagine and
explore
and reconceive right but at the same
time you need to pass back everything
that's recons reconceptualized explored
and

imagined these philosophical uh
modes of inquiry can then return to the
scientific realm so it's nice back and
forth at least that's
that's how i see it and then and like
mel said there's no clear demarcation
yeah i don't also don't think there's a
clear demarcation it's simply
for me the gradient of how well grounded
it is in the empirical
uh roots of these inferential processes
how much weight does it have
um where the other extreme of the
spectrum would be that
you generate a lot of understanding or
sense or reduced
uncertainty by virtue of this reflexive
engagement
with your knowledge i think ryan smith
had a paper on model
uh expansion and elaboration so maybe
that's also a nice connection that makes
sense cool
but yes um you use them
sorry say again who that was that had an
idea
you just right at the end oh right yeah
yeah i think ryan smith has something on
model expansion and stuff
um but i'm not sure um but also you can
also put the paper on abductive
inference and curiosity and insight
which is actually the paper i
co-authored with the
pearl in 2017 um yeah but that's kind of
what i really always like abductive
influence it's
well yeah inductive reasoning is when

you go from the particular
to the general and then deductive is
when you go from the general or the
philosophical idea to the specific
and then abductive is uh when you go
sort of laterally like your your
abductor or something like that it's off
to the side
and so that's this imaginative or
generative way
that we're talking about as kind of
threading between
science with the exploit and philosophy
with the explorer
are we ever going to be trapped into
this dialectic philosophy's broadening
our horizon
and science is narrowing them and then
it's almost like there's this abductive
sideway that we can take where there's a
fruitful
discussion and so that's kind of the uh
niche and the intersection we're
reimagining
so stephen and then anyone else who
raises their hand
steven then mel yeah
i think that the actually that's a very
good point about abduction and
in a way you have to say everything that
we know started with abduction
from a child because it's only with time
that you start to set
um i i often think with induction is
also
you've got a target and you try and
narrow the gap you know
so you but you've got a target so

where's the target come from how did you
know what the target was
um and um
you've also got that question then of so
i think this is
these these questions about how they get
applied
are very very relevant to
um to bringing in this question of
philosophy
you know are you applying it to help
scientists
or are you applying the philosophy to
help
you know you decide to sit and how am i
going to sit in church or how am i going
to spiritually be in myself
so you've got this kind of and i think
this is where
you're starting to see that the same
philosophical questions being addressed
with the same framework
as opposed to it being like okay
that's for them you know it's starting
to break down some of the silos i think
of how
it hits the ground and i think that's
quite important
yeah now
yeah it's sort of sort of what mark
grove said but also sort of
something you said in the beginning
which is um i guess
dan you said this um which is if the map
is not the territory what isn't that
right
if the if the math is not deterred for
what is in that um

uh and i think there's there's a kind of
um classical position in
in i guess both science um and
philosophy
maybe even also the the popular
conception of science um
which is very prevalent in modern times
which is that there are
there are like a finite number of atoms
in the universe
and you know we're we're
picking away at um

[Music]

psychology and we're picking away at the
social sciences and we're picking away
at biology and chemistry
and and so on and so forth but there is
some some fundamental
level of reality you know whether that's
atoms or or you know
whether that's the quantum level or
whether that's string theory or what
have you
but there's there's some ultimate level
of reality
and eventually
you know we're going to figure out how
the psychology reduces to the biology
how the social reduces to the psychology
how the biology reduces to the chemistry
and physics right
and then and then we'll be done and i
think that this is
this is a very um mistaken outdated view
of
what um what reality is like but more
more so what are our kind of epistemic
acquaintance with reality

it's like i think it's it is as as mark
was saying far more far more
degenerative right where we're um
we're instead of instead of uh instead
of
kind of constantly reducing
the the allowable frames of reference
for for understanding the world
we're actually um constantly generating
them
you know we we are we should be doing
right something like model selection
right
some models some theories some accounts
and the sciences
do similar work to others and are not as
good
and those get weeded out right but we're
also constantly coming up with new
approaches because any frame of
reference
right any any model any theory that we
take
is not purely a reflection of
the world out there but will also be a
reflection of our kind of epistemic aims
will also be a reflection of of what we
want to get
out of out of that way of relating to
the world
and so there there you can see how
there's gonna be you know a kind of
infinity of
of possible ways of getting at the world
because they all reflect our
are kind of ultimately subjective aims
as researchers
yes very nice point about the

reductionism is misguided
just the fantasy we're going to keep on
reducing to smaller and smaller pieces
so let's think about the map if someone
says i want a map of california
and someone goes oh no no that's already
been solved it's made up of dirt
see no i don't want that i want to be on
a road trip or i want to be visiting my
friends
so when we reintroduce action and
pragmatism and utility
the argument against reductionism is
very obvious which is okay it may be
that as a substrate
but that's not actually how we're moving
forward
and then it's interesting that there's a
computer science algorithm called
map reduce now it's a different map in a
different reduce but it's almost like
action and inference we map we do sense
making and then we reduce
the scope so we map and we we sketch out
in very broad strokes and then we have
to go into that reductionist mode
to go into a little bit more detail
about what is actually
on the road or in the ocean ahead but
we're mapping and reducing or acting and
inferring
with an eye towards the direction that
we're already moving in
not just because we're gonna uh get to
the bottom
of action or the bottom of inference and
then just call it a day
like just like you said it's an infinite

game we're gonna keep on coming up with
questions
and keep on coming up with measurements
for the dialogue between scientists and
philosophers and ultimately just
participants in the knowledge enterprise
whatever
somebody's title or profession happens
to be so
marco than anyone else
thanks uh daniel that was very beautiful
it's it's the image evoked for me is
like uh
a directed cycles of contraction and
expansion
which uh i believe is also very active
in french
but anyways um so so um
about the uh question about the maths
and language and everything so
i my improvised pitch is
the maths or the mathematical model of
active influence effectively describes
how the terrain maps itself so if
terrain is the
thing itself or the actual reality of
the brain or any ecosystem
then active uniform is basically saying
well all the constituents of this
ecosystem or brain for example
is constantly deferring each other it's
constantly engaging in mutual and
reflexive inference
and all these inferences will then lead
to models of each other and effectively
they all
do this kind of subjective mapping right
but all these subjective maps

over skills and levels then also need to
calibrate to each other and so
yeah if we have to play with words i
would say the maths describe how the
terrain maps itself which i think is
you know quite nice um also uh
it's very strongly echo uh what mel said
i was
very briefly worried that that she would
actually argue for reductionism
i should know better of course um yes so
so
if it's appreciated um i really like
thinking in terms of uh
trees so there's a book by uh
julio africa sorry but my favorite
phrase is um
trees of the brain roots of the mind uh
for someone some reason that always
stuck with me and and basically it's
because there's these generative trees
that
lead to roots at yet another level and
so
um again kind to a
uh the previous discussion uh so the
reduction in story is more about
you know sort of essentialism properties
and objects and kinds which are not a
phenomenon
um but complementing that kind of uh
view
is perceptualism and so i really again
think that this generative
perspective is in spirit with
perceptualism
is how how do these processes actually
structuralize themselves

right how do these flows and dynamics
attain
some kind of regular structure and then
that regular structure will be in the
form of flows and
generative relations um so it's like
generative
scaffolds and i think that would be an
interesting angle to approach anthology
as well
cool oh oh sorry and one more one more
points if you can is
i also want to echo what mel said about
pluralism right so i think
looking at active infants in my opinion
uh i pray enough to talk about knowledge
because then you get the baggage of
correspondence uh notions of truth and i
i think that's distracting if you just
look at it from an active inferential
perspective actually all knowledge the
value of knowledge is going to be
contextual it's going to be contingent
upon the actual agents
right even though we can say oh it's
just the accuracy of the states in the
world as
as believer as mass but these states
don't have to conform to some
objective truth it's more about given
the states that it has models
if it acts based on these states how
accurate would it be and then there's a
bit of a i guess a pragmatist account
of truth maybe i don't know um
but i guess what i'm saying is that that
that
kind of implies also for epistemology

that it's about the normative
contextuality
and that we really should pursue um
pluralism in science
which yeah cool thanks marco we'll go
dave then mel
so i can't hear you dave but you're
unmuted so maybe try to reload or i'm
not sure what's going on there but
mel go ahead
or do you hear dave i don't hear him
yeah
i don't okay so yeah mel go for it but i
know that you can hear us steve but
we're on a live stream so we can't
figure it out at this moment so mel go
for it
um i uh yeah i just want to say
something effective um
i think that that what i guess we're all
sort of
converging to describe that um it
fits in very very well with the kind of
bayesian view
at least at least uh the bayesian view
of science in the future
there are some kind of study old
bayesian views of
of scientific updating um of
the late 20th century
mid late 20th century that um i think
we've
we've surpassed but but i think
if we view um
scientists as kind of subjective
bayesian agents coming up with like
different
kind of parallel models of the world and

adding evidence to these and kind of
retiring
models that don't that don't um
that aren't useful in a way um
i think this is this is very much what
we're describing there
yep it's about these metaphors and we're
talking about
understanding the relationship of
science and non-science that's the
demarcation
project and we've heard a couple of
different metaphors in this introductory
section maybe dave if you want to reload
and try again but
in this introductory section we talked
about one was like a spectator model
with the scientists doing the work and
then there's the peanut gallery
and the philosophers are either cleaning
up behind them
or are confusing in their trail and that
was the following after
science metaphor and we talked about how
that was inadequate
because there's actually a really
fruitful bi-directional
generative relationship and that reminds
me a little bit of a live stream
so it's almost like in the context of
science
uh on the live stream the people who are
on the panel are the scientists
the observable states being emitted
being viewable to the world the
empirical states are like the state of
the live stream
and then we have all of these curious

philosophers
whatever their title or position is in
the live chat
and they're commenting asking questions
and by doing so engaging in this real
time dialogue
so we actually have this instantaneous
system
where we can talk about how the
observable states are in feedback with
what people are saying
at that moment so it's sort of like
again it's just a metaphor it's not that
it carries every single dimension of the
metaphor through but it's just really
interesting that there's a
discussion that can be held amongst many
people in the live chat
that's kind of like the philosophy
literature and then
the people who are in the live stream
could just not be looking at the live
chat
and that would be like a scientist who
just tunes out and chooses not to engage
with that discourse
or the people on the live stream might
be obsessed with what is happening in
the live chat and that could actually
distract them and
help them or prevent them from carrying
out the aims of the stream
so it's a pretty fun way that our new
technological affordances
and the rapidity of them are helping us
understand
the relationships and the bi-directional
dialogue between those who have for

example

technical expertise and those who are
just learning like we've seen on these
streams and in the model streams
so dave let's see if we can hear you and
then

uh yep how's that yep go for it and then
sarah okay

yeah i'm glad you uh mentioned a road
trip

that um reminds me of my puzzlement why
people are talking about homeostasis
rather than

allostasis it seems we've got the you
know and this is a puzzle going back to
freud's

uh proposal for a sign project for a
scientific psychology

we want to get to nirvana we want to
stop everything we want to be
perfectly happy in the eternal now but
then something happens and it's horrible
uh that's preposterous that's not how
any organism ever acts

organisms want to go through a series of
states we want to be

tired so that we can rest and we want to
rest so we can run around and get tired
and be a little hungry so that we can
enjoy the food more we want to follow
trajectories we want to take road trips

um was the notion of allostasis

proven to be useless or to just become
unpopular because

homeostasis seems to fit with somebody's
mathematical formalisms

i'll give a quick response on that and
then we'll go sarah marco

so homeostasis is like the first pass
absolute value you say your temperature
returns to
a certain number so it's homeostatic
it's
thermostatic you know it's just like a
thermostat and so at the first pass
somebody who's looking at just the
absolute value of temperature
or a blood sugar might note that it
returns to a certain point
and then in a static conception of the
universe rather than this process driven
understanding
it's like right if it's returning to
this value of 10 then
it's getting homeostatically attracted
back to 10. but you're taking a look at
the process
and the derivatives the rates of change
and you're like wait it never stops
in fact it couldn't and it shouldn't
stop so why don't we just
focus on the fact that it's doing this
uh iterate road trip
that's yes it's returning to 10 a bunch
of times but shouldn't we highlight the
fact that it's on a circular road trip
rather than just say yeah it's at 10. so
it's really uh two sides the same coin
whether one chooses to highlight the
instantaneous
empirical observation or highlight the
process
and the approach that the system is
actually taking to result in those
observations
so just like many other things we're

looking beyond the superficial
to the generative and to the underlying
models and that's what is
revealing all of these patterns across
systems
so sarah and then marco
um i'm i'm stuck on something math-wise
uh but i think it you know you could
describe it as conceptual
as well um you know when i look at uh as
far as i've gotten anyway with active
instruments and fep
you know it's all predicated on
bayesianism and bayesianism seems to
have a kind of
helios in the sense that you do priors
and you go forward
but um uh i don't know i kind of bump up
against like well wait a minute but what
about
like you know built into fvp there's
like a feedback
type mechanism but it's kind of built on
a substrate that is going forward
and i find that kind of interesting like
i wonder about the limitations around
that
um and i'm not uh well enough diverse to
like
to like say anything more specific but
another little
side data point um which hopefully isn't
a distraction from the question
is you know you you see a lot now in in
descriptions of complexity they'll talk
about well
you you need something that's of high
dimensionality and that's

that description is very much predicated
on the methods that we use
to to to get those dimensions you know
it's predicated on the math
um techniques and i i get uh
even when i think about this high
dimensionality thing i keep coming back
to this like well yeah but you've got
this like
you know f as a function of x or $x y z$
you know like it's not
there's no feedback in the in the actual
um
heuristic well no there isn't a
heuristic in in the end
result or in the in the end um
and you know like fep like let's say it
converges it doesn't
quite converge like markup blankets to
or like the chain
mcmc does but i don't know you get to
something that's close by energy
minimization
but in the end you get to a thing where
you're like where you like
something is a function of da da da and
it never goes back around
it okay that's just i hope that's a
useful question
if anybody has any thoughts around it
marco
and then if anyone wants to raise their
hand to speak to that
yeah that was very interesting um uh
wait
oh yeah well daniel already did most of
it but
i also want to note that um it's also

really what sarah's talking about so
there's this
issue of um having too much freedom in
how you characterize the system right
it's it's not really um well defined how
you're going to define a state space
especially when we're talking about
a complex set of systems that model the
environment so then we have to not talk
about simply physical observables
but but more the structures that they
constitute so for example with
homeostasis and alastairs it's like
daniel said there's two sides of the
same coin
and it needs to be in reference to
something else so you can say
homeostasis in terms of
one perfect temperature and just try to
stay there
and then add to that uh next to it
the processes of anticipation which then
gets you stasis
but another way to also look at it is
for example you can also say
instead of having one perfect state or
a value and also talk about viable sets
for example the domain or boundary in
which you in which the agents
will survive or remain viable and so you
can say okay then that is the state
being in the boundary being in a viable
sense then it reduces to one point again
um which again relates to i think what
what sarah's talking about
i think um was a lot i think very
interesting
um so about the function thing right so

so for example you can have
a a a of blankets it's enacting certain
functions
and it's not built in the loop the loop
is not built into the function but the
function enacts a process right
it generates perturbation activities
flows whatever
that expresses into the environment with
the expectation that it expresses itself
back to them
and so that's left implicit i'm not sure
if that makes sense also you said
something about
um things moving forward and i'm not
sure what you meant about it but it did
feel like it
a very interesting angle you were
pursuing so if you could elaborate it
out i mean
it was it was just really simply like it
was just you know you have this idea of
prior so there's a kind of a teleos
you're like going forward in the way
you minimize energy or whatever
oh so well personally i feel like the
forward
metaphor that then is the problem right
it's not necessarily forward
it's it's optimizing or
calibrating or tuning um but i think
maybe i'm just still an understanding
um but but it's a constant yeah you're
right
that's just that wasn't the right i
agree yeah the moving forward
is um related to historicity and to
developmental thinking so understanding

that
although there might be attributes like
body temperature that are returned to so
that's the homeostasis side of the coin
in another sense there is no state that
the organism returns to
so it's about how will we understand as
you pointed the teleos the end well
there's this local
end directedness as the prior is updated
but then
we're almost using this short-ended
telios because
we don't know the ultimate end game and
as you point out it doesn't necessarily
have to converge it just has to be the
most effective model in that ensemble in
that niche at that time
and that's who does well on that
statistics test
or in that you know ocean ecosystem so
that's sort of this
updating model and um yeah mel and then
i'll say something about the
dimensionality
yeah and i think i think the
active inference 300 principle kind of
framework tends to emphasize
i don't i don't know that this is
speaking exactly
to the questions that you're going
through but um
there's a there's a way in which it it
kind of
looks to to kind of hierarchical
architecture at the exclusion of of
parallel architecture um we don't
we we don't often talk about like the

jockeying of
beliefs at the same level right is there
any
interaction between kind of like models
that exist
in in parallel that kind of thing um and
i think that there are
plenty of you know cognitive phenomena
that get mixed that way
or or um in in biology right
in in organismal biology you know we
miss stuff
because um the fep doesn't
um kind of readily accommodate that as
i as i have come to understand it and
use it
um i don't think it's very good at
handling that and so that's
that's one way in which you're sort of
constraining
uh how highly conceived of kind of
um epidemic agents engaging with the
world
um and and there seems to be a sort of
like
i guess i guess you could say that there
seems to be a directionality
to it that is um maybe misleading
that's that's the time hypercube fep
question with like the gradient descent
and the moving forward through time
what if it's kind of ironically circular
and that
we're experiencing it in the downhill
direction but the hill is just there
and we're moving through it
experientially in a way that is
prospective and it becomes pathological

and not adaptive when it's not
prospective that would be like
prioritizing
retrospection without an eye towards
anticipation
and those systems fail to exist so we
end up having this like
distilled and extracted understanding of
our past
only partial aspects of it with an
action-oriented view
towards an uncertain future so that's
why fep is such an interesting
framework and as mel just pointed out
it's not like it's a lateral
alternative to how the mitochondria
works or how a protein folds
it's not really a scientific idea at
that level
but again to take it to this initial
claim of the paper
the claim is that the fep can be taken
to belong properly to the domain of
science
so then we're talking about what is in
the domain of science if we're going to
be in
science landia is there only one kind of
model
are there multiple kinds of models are
those models hierarchical are they
hierarchical
how do they depend on one another who is
involved in producing and communicating
about those models
these are some of the very interesting
questions that are almost within the
demarcation problem once you're on one

side the line or the other there's still
more questions again highlighting the
fact that it's going to be an infinite
game
with researchers and philosophers so
marco and then anyone else
yeah i apologize if not actually covered
this i couldn't hear it properly
but um i just wanted to add that at
least in my view
the idea of tails for me is at least i
don't like it because
i i have an allergy for everything that
reeks of essentialism
so telos kind of feels like something
universal universalistic
it's almost absolutely great it's like
what is the kids of this thing
right but that breaks down when we're
talking about entities or systems
that are exactly defined in terms of
their context
right so so
each of these systems embedded in a big
mutually inferring ecosystem
they are at the mercy of their context
right
they don't see the world they only see
the larger worlds
through the directly adjacent systems
with which they stand in an inferential
relationship with
right so for me the enormous principle
of fep is the
theological principle as in like like i
think
dan also notes this that the t loss kind
of is constantly changing

but there's nothing else to say nothing
yeah it's not different
than saying that there is a normative
principle
that is defined in terms of the relation
of the system and its context
and so that's what i find beautiful
right so one of my
key phrases is contextuality um
because what's effectively happening
with these systems is they they're
they're they're enacting inferential
context for me
maybe nice parallel is because because
daniel knows that the livestream
parallel is
a domain of discourse so what we're
constantly doing in a live stream
is we're cultivating or constructing and
acting upon a domain of discourse we
keep adding new elements to it
new perspective new frames and then
domain of discourse expands
but it's the same thing for these
inferential systems these marketing
magnets
right they are constantly changing their
domain of discourse when they reconnect
with other
inferential adjacencies or let's say
interlocutors
they change their domain of discourse
because that domain is defined by the
adjacent systems
at least in my opinion this is again a
bit speculative
away from the brown's truth but i'm
curious to hear what you think

the patterns of discourse define the
interface

so it's not just that there's the
essentialist nodes that are then
like bumper cars on a you know in a chat
room who are engaging in this thing
called discourse

it's like it's a co-construction just
like you pointed out with the relational
and the contextual or the ecological
teleology the purpose is multi-scale
and it's it's relational and then also
teleology is just one of the quadrants
in the two by two with uh aristotle's
four wise the tin bergens four wise
so there's the teleological cause but
also there are the other
quadrants so there's other aspects of
why

that were might be interested in and at
any given moment the why
that's interesting or meaningful um is
itself situational but there's also a
pluralism

there so it's kind of like we're weaving
these similar
ideas of relational thinking pluralistic
thinking

and then we're finding that we're kind
of pluralistic across
issues like we're okay with people
having different perspectives in a live
stream

we want to have a plurality in the
literature but then at the same time
that's that
expand cycle and then there has to be
the reduce

cycle because if we have pluralism that leads to somebody thinking that we should capsize our ship it's not that it's uh you have to do it on all possible universes this is like simply wrong it might be enough for someone just to say it's not my preference it's not my c vector that this ship get capsized so that's not a policy i want to engage in so it's kind of like this moderated pluralism and this uh pragmatically oriented dialogue with philosophy and science stephen and then anyone else and that's this pragmatism and being able to move between scales the the idea and it talks in the paper you know about thingness the idea of the something being a thing even if it's not a living thing you've got that and i'm curious it didn't actually quite doesn't actually quote the the markovian monism sort of tag in there as that so i'll be curious if that that was a particular choice they didn't get locked in by that um but um this this this idea of dimensionality is if the free energy principle or is based on entropic change and entropic is is dimensionless to some extent it's just a scalable

eigenvector sort of
um i don't know what you'd even call it
quality
and then it goes into thing this then it
goes into
some sort of process theory which
has to have dimensions at some point you
know so
you know we want you to in some ways
once you go into active inference and
there has to be
dynamics across things you start to have
dimensions or maybe certainly once you
have
something that's nested inside something
else and that's
that um i just think that's really
interesting i'll be curious what mel
what male's thoughts are about how the
dimensionality
drops in and out between thingness and
say pure entropy
and um and and just maybe why she didn't
put in markovian monism
was that a choice for um
sort of reasons of not getting caught up
in it or does she not agree with it or
something like that could you just
briefly introduce what markovian
modernism is or why is it exciting or
relevant to you okay so
well markovian monisms um is the kind of
overarching overarching philosophical
um uh
sort of uh paradigm where
the um the the the idea of
something being in the world is because
there's

there is a differentiation between
what's outside and what's inside
and therefore there's a sense and and
the nature
of how that inside and outside dynamic
is structured
even be it a stone in terms of the
stones stays as being the stones that
somehow the forces inside balance
its relationship to the environment
outside and therefore
there isn't um there isn't this dualism
of
the stone and purely outside the stone
as a complete separation there's this
somehow
um emerging so everything is
to my understanding this is is monism is
there's
kind of a an underlying kind of
unifying kind of uh
dynamic that we um which sort of builds
on sort of ideas of david bone
and the kind of um that those sort of
principles that there's something in the
background which is
um more um implicit
in our existence if i've said it right
but that's my understanding
okay it's definitely a uh you know big
and deserving
topic usually the monism would mean that
there's basically one kind of thing
dualism would be the idea that there's
perhaps two kinds of things
so the one type of thing in monism
whether it's matter or whether it's mind
is being manifested differentially

these two views often converge because
the monus will say yes but it looks like
two things or the duels will say yes but
they're interconvertible or they
associate with each other
so it's a little bit of a weedy area
so if there's any thoughts on that where
you can happily hear that
otherwise let's um maybe slowly walk
through
some of these slides and just highlight
some of the very
nice points of the work itself and then
just think about how that kind of
springboards us
into 2021 and beyond why not
sound good yep all right cool yeah
go first and then we'll continue on i
could yeah i could respond quickly um
so there there's a paper um
a few years ago that presented um
one view on consciousness that that carl
christian was a uh an author on um
and i think the the phrase that stuck
was something about the cartesian
theater
so it was it was very much a um
kind of classical duelist uh account
um and uh this more recent
one right this is this is just this past
year right
um the the markovian modernism one yeah
so
it's another it's an updated view on
it's an updated take on
on the consciousness problem on the one
body problem um
that question is also an author on um

i think it's interesting it's an
interesting um
they're doing interesting stuff for the
formalism with the information geology
stuff um i used to be very
um compelled by
the the problem of consciousness but
then my body problem i was very um
i worked with that as my undergrad that
was my uh so that was
that was a big thing for me right now i
was like interested in evolution and
interested in consciousness
a couple years away from the mind body
problem before i come back to it so i'm
trying to like
i try to like keep my blinders on and
focus on
on like one or two problems at once
yep philosophy has that feature of you
pull on one question and then all of a
sudden you're asking how do i know
and who am i what are we it's because
they're all
very linked up and kind of like it could
be the same with science you make a
measurement about the gazelle and then
you want to know about the cell
but in philosophy it's almost hitting
you all at once
and there's less clarity about how to
structure those questions
and that's actually part of what i think
these conversations help clarify because
in the example of the animal we know
that there's a certain nesting
or interactions between different types
of biological entities

and so we can coarse grain and that
allows us to make
useful decisions like about wildlife
management or something like that
again with an end in mind and a team and
a method in mind
we can do useful science and so the
question is with the method in the team
and the question
how could we do useful philosophy and
then maybe even move beyond these
titles overall and just have the group
working on the project with the approach
so marco then mel
i just briefly wanted to uh uh joining
the social unconsciousness
because i'm not out of it yet i'm still
obsessed by it um
so so so i think the beauty of
microphone is or
or dual aspect mode uh is is
because the most beautiful thing for me
for active inference is
the emphasis and the primacy of
mutualism
right that nothing is without context
nothing is seen in context
um and that all their so-called identity
so-called properties so-called t-loss
these are just heuristics we've
accumulated over the years in
ignorance of the relational nature of
these systems
and so the beauty is that these
relations
of relations of relations these tangles
of relations themselves
become thing-like because they

facilitate or allow for
instantiation of a coherent activity i
forgot what the papers i've been trying
to find it's
i privacy dig on twitter but there's a
very nice paper about how some nerve
populations that might for example
hypothetically
are about encoding cats this just put it
very simply
and they're just constantly chaotic
until they get an input
that is relevant or comes from sent data
about
cats and then that chaotic high
dimensional
noisy stuff collapses into a particular
order
um and i think it's also interesting for
you stephen because you also think about
spaces
and and this relates to a lot of well
some people think it's too hyped but the
work on manifolds
and computational manifolds in
neuroscience so
the weird thing is you have this really
uh you know a lot of neurons and
population
and they're related to a relatively
challenging
imperative problem but then for some
reason when you analyze the data
um you can obtain a manifold you just
say well the neural population
is basically just traveling on this low
dimensional manifold so that's very
interesting

the interpretation of that is another
topic but but
since you asked about the nationality
i'll just add that
yeah actually mel
yeah it's what what you're saying just
reminded me
i had a thought yesterday i was i was
trying to
nail down one thing and of course you
know there's this
right you you pull at one thread and
everything unravels um
um i i see that there there can be a
kind of systematic way of going but i
think
i think philosophy is good it's not so
chaotic
it's it's right it's it's using the same
kinds of
of of conceptual tools
just not with the formal rigor of
mathematics but but there's still kind
of a
uh system it's just so it's sort of like
you know a mean field approximation or
um solving a system of of
differential equations approximately
right where you're
you're um you're you're taking you've
got you know these 11 moving pieces and
you're taking 10 of them
and your your either sampling over them
to get an
average value or you're you're
approximating them and you're you're
kind of like
freezing you know you've got all these

dynamic moving pieces but you're

freezing the values

you know so so to do this paper right um

there's no one definition of what a

model is

right in philosophy of science there are

you know

50 people working on models and they

have you know

at least eight different like distinct

conceptions of what a model is and

and what a model is rests on what

similarity is what representation is

what analogy is right and when it and

then you you know you dig into these and

they're all these different conceptions

of what similarity is and what

representation is and what analogy is

right

um we don't know exactly what a theory

is we don't know exactly what a

uh the scientific method is right

but to to drill down on one thing

you've got to kind of stipulate you've

got to kind of like average over the

literature

and find what it is that's that's a kind

of commonality and what we think about a

model

and kind of stipulate well you know no

one no one agrees on exactly what a

model is but a model is approximately

this thing and in order to figure out

you know what this one model is doing we

can sort of stipulate

roughly what a model is um

and it it connects

quite similar i think to what we're

doing in math and physics
but um just using language
yes um mel i think you've taught me more
in that like whatever three minutes than
i learned from all my professors so far
so thanks yay that's fun stuff i agree
that was really
awesome and again it's like philosophy
and science we're almost
converging on this approach if not just
converging on the conversation because
when you were talking about how many
definitions of models there were i
thought about how many definitions of
foraging
there were for insects or definitions
that go beyond insects
and then i thought about well when we
measured it in the field we said it has
to be
walking in a straight line with
mandibles empty
between this time of day and this time
of day and there's no single point that
we would draw the line in the sand is it
one meter is it three meters
it's always going to be like you said
it's about looking at the distribution
of the literature
if it's outside of the bounds of
literature you're going to have a lot of
justifying to do to the reviewers that
doesn't mean it's wrong
it just means that you're going to be
having to make special claims like
you're going to have to write
an extra subsection in your paper to
justify why you did that

um but then if you use exactly what the literature did again whether it's right or wrong it doesn't ask to be justified and so people who are operating within that center of that mean field approximation they kind of just point outside and they don't need to do an internal justification of their choices or of their approach and then you just said like you have to freeze other things so you did the scholarship to show there was this whole section on scientific modeling and then all you can really do is trace out the if then you said if you are feeling this way about models then you would be consistent in thinking this and it's not that it's better to be consistent versus have cognitive dissonance it's just that you can do the scholarship and trace out the if-thens of thought so it's just a fun way because there's probably people who are hopefully listening to this and curious about the conversation who are coming from the philosophical side or from a scientific or from an engineering background and so it's like there are similarities across all of our experiences like the need to iterate the need to freeze specific subparts of a model whether it's qualitative or quantitative and then work with teams and with

literature

corpuses that aren't just giving a
single resounding answer for what is a
model or what is a foraging trip
so really just important nomadic ideas
that we're

broaching here so it's just good to take
stock of that so marco and then we're
gonna just walk through a few of these
slides

unmute marco

um yeah i was saying you you've again
relieved me of a lot of stuff i wanted
to say

um but but yeah i also echoed this um
i think what we're seeing also in
philosophy and

science is moving away from or moving
towards pluralism

and i don't i don't think it's a problem
that there's no consensus on the single
model i mean the model is not a natural
client it's a concept we we're using
because

our again non-natural or abstract
processes of inquiry necessitated the
phrase

model and so a model itself is
contextual

or context dependent so it's kind of a
fear relatedness

or context lateness or normal bladeness

i guess

but i also like the the comments of now
systematicity right so

we've inevitably accumulated all these
conceptual tools

and inevitably people will try to find

system testing between them
and i'm kind of wondering uh in a weird
reflective way if activision also can
shed light or uh make interesting uh
that phenomenon of
of what we constantly said uh letting
the fly out of the flyball
we've surrounded ourselves with so many
tools bonded together
with systematicity without actual real
justification
uh and so we're kind of trapped in that
fly bottle because we haven't had
an alternative so you know who knows
active instruments can
fulfill we can stand stream so on the
fly bottle
um in undergrad i worked in a fruit fly
genetics lab and the joke was you'd have
to fly in the bottle and you'd say
there's the fly in its natural
environment because it couldn't live
outside the lab it would get selected
out so it's like no we're studying it in
its natural
environment now that was laboratory
biologists
who were just very aware of how
selection works but it's such a deep
insight into like yeah you are always
studying it in its setting
whatever it is and then you just
broached upon this contextual and
relational thinking
and then also that modeling it's not
just about the object or the essence or
the natural kind
model is a verb so it's like the

dynamical

and the relational approach is coming

and imbuing all of the things that we're

working on so steven's working on

performance

mel is working on the history of thought

and about the discourse that's happening

at a very

high level we're all working on

different attributes here

and then there's something that we can

align around like a strange attractor

itself

and that's that we should probably be

thinking about context relationships

dynamicism multi-scale models so that's

what crosses systems and that's really

the

fun conversation that gets to emerge in

these moments

good i'm glad that people got riled up

by that so let's go

stephen then dave then mel

one thing that also comes to mind i

think that's quite pertinent maybe in

the broader

context which everyone the universities

you see it this interdisciplinarity

transdisciplinarity participation

um mostly

it's words that the universities are

getting into because they

they start but that it raises massive

questions because

um how do you start to not just try and

contest

an idea within a field or you need to

work between fields across fields

you need to bring in context and
multiple contexts
and i think this you know question about
how we actually
start to understand knowing and at fuzzy
levels of knowing
which actually opens up potential to
bring in people who are more
marginalized
because they have other ways of knowing
and experiencing you know and
they are better adapted probably to
survive in some very difficult
conditions
compared to the the research themselves
you know
so sort of without making the you know
the genetic
comparison but you know the the whole
ecosystem and bringing it all together
so i
i just wanted to bring up that question
of like how all of this
does relate to transdisciplinarity um
and the need to do that with the world
in the state that it is
okay chill dave and then mel and then
we're gonna move through sight
yuck panks up and made a comment along
the lines of your fly bottle
um he found he could not work with white
rats these
specially bred rats when he's
researching the evolution
and the socialization of emotion because
these are completely artificial
creatures
with no normal rat emotions you can't do

those things to a
a real rat that
rat runners do with white rats what he
took was
your regular old brown
alley rats the nasty horrible rats that
everybody's afraid of
raised them around people for a couple
of generations was really careful not to
select
out anything and they just got to loving
people that could play with you and come
over to be tickled and they'd giggle and
have a wonderful time
and he says you know these are real
things these are actual animals that
could that survive in the wild i could
let them go and they'd go out and live
the
white rats would be just wiped out
overnight
okay fair to say that yes it's not about
the exterior
shades in any species mel on anything
else you want to say there
i actually forget what what i was going
to understand but
yes chill it is very interesting and
it's always the real
environment and who's adaptive and who's
not it's it's not something that can be
speculated before the ecosystem emerges
because most species go extinct and
there's insects that
had a strategy for 100 million years and
then they went extinct and it was always
situational and
interesting questions though so we're

gonna pass over the
abstract and also there's about half an
hour left
in case people want to put any questions
in the chat
so in the roadmap i think it's a good
place to pause on our road trip
analogy we can start a we can see some
of these terms
that we've brought up in the discussion
reflected
by sections in the paper so um and by
the way if anyone has to leave that's
always chill um mel go ahead
i have to leave yes thank you i saw i
saw the time drawing
near mel thanks so much for
participating in 14.1 and two
it was just obviously great discussions
and you know
apologies for bashing you after you
leave no thanks for
coming on oh please do yeah in any other
future discussion you're always welcome
to contribute your perspective so we'll
hope to see you soon
fantastic thank you cool
so nice we can all you know continue to
go around but really interesting stuff
we did hear a lot of these
things let's look at this table and
maybe are there any terms that we
haven't really brought up we've talked
about
for example i'll mention mean field
i can see a few things already but maybe
someone can raise their hand and pick a
term that

they don't see how it's fitting into the things we've already brought up because these discussions it's like it's not the end-all be-all it's just a couple of hours mel probably spent hundreds of hours writing the paper so we're just sharing our perspective seeing at the first pass how these ideas are associated and then there's a paper and a literature and a research community to delve into if some connection is interesting so someone's like well with the connection between 6.1 and 4.3 was really exciting so dave what would be something cool to go into yeah um i first encountered the term free energy in economics where it means something just dramatically different it's a good thing in economics um but the closest that i've seen to a use with the connotation that that we've been using is as anxiety free floating anxiety paralyzing terror then i don't know what's wrong is that uh off the wall this is fun um it's definitely an unconventional economics interpretation of free energy but it just shows how when using terms it's always important that we understand where we're coming from because in a way that's what this paper and also several other people papers have um

brushed upon which is there's a
thermodynamic aspect
to free energy gibbs free energy which
is like
after you ignore the activation energy
so whether or not the kinetics of the
reaction are such that it does
go down after the reaction has occurred
delta g change in free energy is like
how exothermic the reaction was is what
draws the reaction forward
and so you have a candle it's like water
in a dam
it isn't spilling over and then the
delta g is negative so it wants to burn
and then when the activation energy
occurs it does burn so that's the
thermodynamics
free energy there's more to say there
and that is actually
addressed in mel's section on
statistical and thermodynamics
the origins of this free energy term in
this setting
are from again statistical and
thermodynamics
but as that became more and more
integrated with information theory
what's called the epistemic turn there
became more and more of a marriage
between physical interpretations of free
energy
and more informational or statistical
and even bayesian
and computational understandings of free
energy so then
we were at this point where free energy
was being used pretty fluidly as a

metaphor in machine learning
by analogy to those chemical reactions
but now in the statistical setting
and that set the stage for the
innovations that fristen specifically
made which we can go into
but that i hope was a little recap it
sounds like economy
drew this free energy term and then just
to close on that you said a generalized
anxiety
yes it's the differential between um
you know being too anxious and having
too much surprise
and being too um rigid and fragile
though converging to a perfect
optimistic model is not a bad thing
so it's sort of a gradient descent on
expected free energy
because just the reality of the
situation is that we're not going to
come to that perfect model really in any
meaningful respect
it's just not how but these are
there's other you know little pitfalls
and land landmines and things and these
are all
papers that are in a network like the
paper that we're going to be reading in
15 about
is the fep realist or instrumentalist is
it actually how things are
or is it just a scientific instrument so
those debates are being
skipped over in this section but that's
a little bit how free energy got from
statistics um and statistical thermo to
information theory

marco

thanks um uh just fair warning i'm gonna
be a bit expected

here but but um i always like when
people uh

make nice metaphorical connections so so
they've um yeah i i i would want to um
um build upon your your intuition so
so um like mentioned we have
thermodynamic free energy
and no matter what these thermodynamic
free energy can be used for work
right but then we can ask okay is the
work that's going to be done
is it going to be useful or not for the
agent

and so this ongoing process now we're
just talking about thermodynamic energy
this ongoing process of basically
channeling or or allocating or
distributing the thermodynamic free
energy

in a way maximally useful for the agents
is a challenge so for example it can try
to challenge somewhere

which will evoke a certain inferential
challenge but when it fails

you'll get negative effect right so so
so one part of you

sends a challenge the other part can't
resolve it and then you feel it so

so yeah i i i kind of feel like it's not
that wrong in my opinion

to to see that because anxiety is very
uh specific

it's it's basically kind of like fear

but it has no

targets right it's generalized so you're

afraid of spiders

but you have social anxiety because the
whole domain of social interaction is
just fraught with uncertainty

but without the availability of a
specific reason why

uh your you're negatively uncertain
or in an uncertain negative way that
makes sense

marco i'm going to build on that because
it was really helpful so
you mentioned how the fep has to be
deflated

because it's like generalized bad it's
not the same is bad but it's
something that is being optimized
whether you think of it as minimizing or
maximizing depending on whether there's
a negative sign so it's the kind of
thing that we're working on

that's the key point and it's inference
what we saw here is that we're going to
draw a physical analogy

from the way that chemical reactions
indubitably go downhill

the candle doesn't unburn so we want to
design

governance structures information
schemes economies that are like a candle
and they just irreversibly according to
our experience they just play out you
don't have to specify the micro level
detail

just as an analogy so we want something
that's going to go quote

downhill in governance space the thing
is we don't just want to simply do
inference on it so in the machine

learning area it'd be like okay what's
the data set let's fit the estimator
let's reduce free energy so that we can
fit the estimator on the descriptive
model

like on a big k-means model how many
clusters are there then that will be
like

an inference model we're actually doing
free energy gradient descent on what
policy selection that's where control
theory and cybernetics comes in
because we're not just doing descriptive
analysis of parameters

of large data sets we're actually doing
prospective or

anticipatory policy selection in light
of a deep knowledge

reflected by embodied priors in a niche
so that is the oriented turn where we go
from again just pure

free energy minimization to do inference
via

tricks and heuristics like variational
base and fristen

takes it to the next level by doing a
few things not the least of which

are incorporating action conceptually
doing inference on policy selection

which we've seen embodied in the models
where like the pi

is influencing the b so the way that the
state estimate is changing through time
the way that s

and the b matrix are related is being
modified by policy

so policy is what free energy is

plugging into so we're doing free energy

descent

on not just descriptive attributes of
the world or any smaller large data set
but we're doing a prospective free
energy grading descent on policy
because that's about our world model and
how the states the world change
for some people i know i was a little
repetitive but that's really
some of the key history and that helps
understand how
statistics and thermodynamics and
bayesian inference which already is a
big

synthesis to make how all those areas in
information theory and computation and
everything that's kind of quantitative
that we're talking about

how did that get involved with
ecological psychology
control theory cybernetics and then even
into this deeper relational way of
thinking

so that's a little bit of that and it's
an intersection that will you know
embroider over many times but it's
really a great story

and it's awesome that mel wrote it up in
a way that isn't

just backroom lore but it's something we
can be like yeah it's in

3.1.1 in mel's paper so that's what's
great about literature

stephen

yeah thanks daniel just got to take a
breath because you that was really uh
that was quite a lot of
useful information i agree with what

you're saying there and i i suppose one thing that is good to bring it back to that gibbs free energy is you know you know there are times when for instance ice when ice freezes and the um the actual um free energy um is such that it gives it acts as a force it acts as a force so that um the the the the lower energy molecules um are not necessarily purely favored and it takes a slightly different structure and you you know you end up forming ice of which um you know it goes to a state which is um not necessarily energetically favorable you know so but that has that that dynamic is still something that can happen over time with equilibrium so you've got this ability as you mentioned to to sort of um to select policies um outside of equilibrium to to um to to keep going and to work with that um but i think that what the free energy does and what this talks about that's a bit different to how a lot of people talk about and stuff is they normally work with these phase shifts and this kind of like if you put enough molecules in a box and get them to be constrained in this way and that it will emerge that you'll get the ice

or emerge that the ice becomes water
um and that seems to be like the
prevalent way of thinking about
emergence sometimes in kind of economics
and some other cases it's just the
interactions
under the the the temperature and
pressure
but what this gives away to think about
how it can be
um innovated upon and thought about in
terms of
um policy selections happening at
multiple scales
and uh the the challenge is it depends
which scale you go at and you're talking
to
to try and explain it to someone you
know so the cell
versus the organ versus the organism but
i think that um that that's something
that this
um really helps to unpack um because you
can't deny that
this is not just emergence as you put
everything together and
they'll interact and it is it involves
some sort
of um sentient uh or some sort of way of
building up but starting from
a non-intentional base so you and it's
got both
which i think is really cool thanks
steven i want to
mention two points so you talked about a
phase change and so that's a reversible
phase change with like melting and
thawing so it's a little bit different

than the candle you know melt thaw
resolve itself into a dew
that kind of stuff so slightly different
with a chemical reaction versus freezing
just from the chemical side
but you mentioned it was not favorable
for the water molecule to
freeze from the liquid into the solid
face so go from freeze from
liquid water into solid water but
actually it is
it's that as the thermal energy reduces
all of a sudden it clicks into a new
differentially configured arrangement
and so that's what the phase
is is like when you're at absolute zero
i assume
most substances are solid like they're
in whatever their ground state
configuration is
which is like a close packing it's a
crystal of some kind it's a solid
and then there's some temperature t
where
that crystal melts where all of a sudden
it's like the iron is vibrating it's a
crystal it's a crystal it's a thousand
degrees it's 2000 degrees i don't know
what the melting temperature is but at
some point
the iron is breaking out of its crystal
and goes into the liquid phase
and then there's still the
inter-attraction of the different
molecules in the liquid even the iron
and then at some point it vaporizes when
you're even hotter than that and plasma
and so on

so these phase changes are actually
they're about the temperature and then
if you want to go to that fallacy you're
kind of mentioning
it's like people think that you're gonna
get a phase change by adding more
into a box or by stabilizing it but it's
like no if you're at the wrong
temperature
you're never gonna get ice you need to
do an enormous amount of pressure
to get solid above zero you have to
crush with like millions of pounds of
pressure but you can make
water into ice at like five degrees
celsius
you just have to put a massive amount of
pressure and so the question is
how do you do it with zero pressure
that's by changing the temperature
and that's where this multi-scale yvonne
in the chat 2282 fahrenheit for iron
thanks for that
super clutch then the second point just
quickly and anyone else then sarah is
which scale and you said the cell or the
person in the society
and to me that's making me think of
helen longino's
book on pluralism because she discusses
how the different fields like the
sociology the geneticists the molecular
biologists
they have stories about aggression and
sexuality in
the case of her book and depending on
which method you're familiar with which
affordances you have at hand

which aspects of the world you're
sensitized to you're going to tell a
different story
a molecular biologist is going to have a
protein binding
answer to for example a pandemic
a sociologist is going to know about
some social intervention
and so there's all these different
people and it's not just that one
person's worldview is like
the right model and someone else's is
the right one that's kind of this
question is which scale
it's not for any one person per se to
say
but those are the questions that we need
to gravitate around
to make action that isn't unduly biased
towards the molecular
towards the social etc but rather
something that works across those levels
so really interesting stuff steven
thanks for like always
highlighting how it comes back to bigger
questions like that
sarah and then marco
yeah i mean maybe just to throw some
more monkey wrench into this
um something about what you said really
really um reminded me of
um the phenomena of hysteresis and i
never really thought about hysteresis
this way before but i was like oh
jesus i mean it is true i think that
hysteresis is an expression of
of hysteresis historicity um you know of
history

and and so um you know
with with fep i'm not sure with active
inference
maybe but like i often ask myself how
these various you know like you have an
entity as
you know by way of your model or natural
kinds or whatever it is but
you know when you hook in these enti
you're kind of you know weaving or
whatever these entities
together um by way of something
and and hysteresis is always been
interesting to me because it's like
that extra bit of dimensionality that
phase if you want to call it
um that can hook into something one of
its neighbors
and so i often think about the um
like let's say with fep or active
inference you know like
if there's only like one interface that
can be used to hook to a neighbor
you know maybe it's phase change or
whatever that that moves out to an
adjacency um so like how many different
degrees can be used to connect as an
interface to various neighboring
agents type thinking and i know that's
like
left field i don't but daniel if anybody
can take it somewhere
you can or marco i don't know marco
definitely take the first
swing at it or anything else you want to
talk about
thanks i'll do my best um so
let me just grab my thoughts before i

lose

oh yeah yeah uh sarah but what you just
said i'm not sure if you find it
relevant but one of my favorite papers
is from um avena konigsberger
and all of spawns and other people about
um

routing strategy so so i i personally am
very fond of
the conceptual set of flows and paths
and routes

right because you know quite selection
can also be seen as navigating
a certain policy space so this paper
spectrum of writing strategies
effectively

is tackling an interesting question of
when you have a given system
how much non-local how much uh
information

in its adjacency its context doesn't
need to incorporate right so as it
expands

let's say its domain of discourse or its
inferential context

as it expands there will be a non-linear
growth in let's say the dimensionality
or the rank with the involved elements
as it were

and that's a cost at the same time if
you don't expand

maybe you're just not able to resolve a
certain differential challenge

and so this trade-off of of how local
and how global or how expansive

you want to go for a certain
differential context um

is i think related what you're saying

right so how many degrees
can you afford uh when it comes to uh
selecting how you go about selecting a
policy um
if i answered your question i'm not sure
it's right but i linked it in the um
youtube comments
thank you um and the other thing what
we're talking before is thermodynamics
oh yeah
i am still not mathematically
um competent enough but for those who
are
i linked also uh in the youtube chats
to um a article by uh john carlos baze
who is
a huge figure in mathematics moments
and he wrote something about entropic
forces and when i
discovered this it was just mind-blowing
and amazing because basically what he's
doing is he's licensing
talking about forces as governed by
entropy so
so it's basically about
whether the temperature is static or
whether it
is either thermostatic or or motion
centered was it
anyways um but when the thermal
the temperature is static then the
forces
are better described in terms of entropy
um
which i wanted to emphasize because we
had this discussion a bit with uh
steven's point earlier and i i often
like saying to people when they don't

like

normative principles there's always at the very least the ultimate normative principle

which is the principle of least action

you do physics

the cycle of thermodynamics there's always that no matter what every system has to be governed by that

so given that we won't deviate from that law we can say the brain is governed by a principle of this action but clearly

it also is governed by uh imperative to systematicity

uh or mutual inference this constant co-regulation

and so the question that i wanted to explore is kind of how does that link together

what how i like to what if you phrase it is basically

these kinds of systems are effectively cultivating nature to do the work for them

so how do you transform the dynamics governed by the uh by the ultimate principle of

fiscal peace action how do you nudge that into

also being systematically conforming not just physics but also

your inferential imperatives if that makes sense

so this is a very subtle and really interesting point

marco thanks a lot let me try to trace a little of that

so you were pointing out how bases work

on entropy and some other aspects of
information theory and the fundamentals
of physics
it's almost like integrating statics and
dynamics into one
more unified field and
you or or you're muted but is that
correct or where
where did that and i i i'm trying to
trace how this
comes into the thermodynamics side
because
continue though i'd like to understand
that a little better
yeah i mean honestly i think we should
have discussion maybe uh
yeah later because it's actually brief
post i'm not sure if he has a formal
paper
about it okay um but in general base has
a lot of interesting work about
entry and biological system again you
know way out of the element there
to be for me but i would really be
curious to see
what what more formally uh competent
people
think about these terms but the thing
about entry force is just
a very simple blog post and it's it's
it's it's not that complicated
okay we'll read it discuss it and maybe
invite him or anyone else to be on a
conversation
so we have about you know 10 20 a little
bit of time left so let's think about
some thoughts questions we're
taking forward with us but then we can

kind of flip through a few more ideas
sarah to your idea of hysteresis and
historiosity which are both two kind of
like
similar sounding words but hysteresis is
just the system where like
when something happens it doesn't return
to exactly the same state
and historiosity is just the idea that
things have a developmental history
like even if you reelect the same person
they're older it's a different world
etc so then you also mentioned the
connectivity
of agents and then that made me think
about social
agents and how the social
networks have a hysteresis element to
them like if there's a fracture it's not
like you just
zip it back up like a sweater it's like
gonna be something that has to be
um recovered if it's recovered exactly
or not exactly it has to be recovered
in a different way so i kind of see
where you're coming from
with the hysteresis and the
connectedness of agents now
where does active inference play into
this
if it if it is related i think it i
think there's some
elements because we're using a lot of
these words so it's kind of like these
networks of
thinking about these topics i'm not sure
where one would take that
going from the idea that it's the

networks and the relational insight
that defined function of the system and
then hysteresis and thermo i'm
thinking of a few things though marco
yeah well i mean i would love to hear
your thoughts about it so i think i
think
there's a bit of a tension because often
the formless is described
in relation to mark of change which is
explicitly about just
being dependent on only the last state
right but i i i think what sarah was
kind of alluding to quite rightly so
is is when you have these uh not just
one system but your interconnected
systems
um then the a
a past it of a particular system might
effectively
come back with a detour support and and
i think that's kind of what she's
pointing at
even though you could of course define
the mark of chain for every point
etc um on a different level you can say
that
the past states for example that system
expressed an influence
into its larger context and then later
another system that received that
influence
uh responded with again a return signal
influence
to that initial system that makes sense
yeah this is actually what i'm always
obsessing about
um is and you can analogize it to

complex numbers as well you know like
the hysteresis adds that it pops out an
extra dimension that something else can
hook on to
um and that's what i'm i'm i'm always
thinking about with respect to
you know um i don't know if not even
well-versed enough to even say fep but
what i understand about active inference
you know like
that the model seems to just creep in a
forward direction but there's no side
chains like there's no
there's no side hooks um and so i'm
always i'm always like yeah but how does
that how do you
do i don't know like i'm it's
and it may just be my limited
understanding but but i come from a
place of thinking a lot about hysteresis
and memorisers and stuff in it
and i can't quite get those the math to
answer my question
did she elaborate on side hooks i think
that would help
agree what would a side hook look like
not a sky hook
for those dennit fans out there well
what i mean by like let's just say you
know like you have real numbers and then
you have complex numbers and i think of
the complex plane as like a side hook
um where you you add an extra dimension
basically where other things can
interact
with that number space by way of the the
complex side hook but
it might be a too limited paradigm for

me like i'm throwing out these
random ideas that may or may not um
work together but that i don't know it's
it's
let's say uh this is like totally
sci-fi um
you know like you have memorisers and
they're in that but they're going in
like voltage amplitude one direction but
then they have
different paths um as they go forward
versus reversed
and and so that that phase that kind of
phase
relationship um might
offer a relationship with something else
that's related to phase or related to um
you know whatever the physical dynamics
might be around that
so i don't know like when i think about
um active inference
you know like they they're just like
well what's the happened in the past and
what's going in the future
what are my prayers but there's nothing
on the side that
um sorry it's as good as i can do
i don't know if that's helpful you know
you rile you riled everyone else
else up so it's great so we're gonna
just slowly we don't need to end exactly
on the hour if anyone
is gonna um leave on the hour that's
fine just um maybe speak your piece as
your closing piece but we'll stay on a
few minutes after
because yes this is a really nice topic
so marco

dave stephen yeah thanks
i think i understand what you mean with
side hoods and i think
maybe what you're what you're um bumping
against
is is what i also find uh problematic so
it's good that you raise this so
so the the the the equations the
formalism for free energy
and uh for fep and active infants
they're standalone they're like the most
naive equations you can have
right but the whole point is that when
we apply then we have to fill in the
things that
were previously naive we have to for
example only now only very recently
we're going to multi-agent systems with
active influence
but that was always implicit in the
research
as in from the first paper in 2009
2010 it was already implicit because it
explicitly talked about
how it would be multi-scale how it would
be um
multi like nested right it's all these
different systems all doing
frenzy minimization in relation to each
other
and so that that codependency right
these mutualistic relations those are
the scientifics
i believe so for example uh if i'm not
abusing maths here you know the decide
for complex numbers is
it's imaginary until you square it right
so it's a potentiality until a certain

operation occurs

so in the same way when a particular system expresses an influence into its context

it isn't affecting that system anymore

until something happens

when it returns right so so the fact

that it's always situated in an extended environment

um that a particular system is very naive about

actually that it can't know everything

exactly that it has to kind of trust or

upload certain

um influences to its uh context

is kind of that side i would say um

does that make sense

interesting i think that once i look at

the math i'll keep that in mind and it

might start to it's

that's really helpful don't don't look

at the math don't look that's the point

right you'll hear v maps

describe a general uh

general principle for how they are by

what they are governed

but they are governed in terms of states

and relations to other systems

right so i i kind of confused because

you've alluded to it many times before

how they're all connected to each other

right so i think that's not in the math

oh wait i'm just going to continue with

it

i think i think it's a little bit in the

math though i think if i keep your

this is how i think or how i learn is i

keep these kinds of phrases and

statements in mind and i look for them
in the math and then it becomes clearer
so i i think
yeah and you know and you've talked also
your metaphors are great you know like
you've talked
also about context and shifts in context
and and
you didn't say this but my brain did
this like collapsing or like something
becomes um so i'm gonna it's really
helpful to keep these things in mind as
i'm
as i'm battling my way through you know
these formalisms
um so yeah i'm just gonna ask daniel
because danny was way smarter than me
i know there's a mathematical relevance
here because of matrices
you have a marginal the diagonal and
then the other entries are for the
conditional
conditioned situations and that also
reminded me of the scientists if that
makes sense oh yeah
we're gonna we're gonna return to side
hooks but let's go with dave
then stephen okay um
hysteresis and side hooks uh the first
the most elaborate discussion i've gone
through on hysteresis is in renee
tom and chris ziemann on morphogenesis
structural stability and catastrophes
in there um the
hysteresis are refer to hidden states or
relatively
inaccessible states so these these
amazing things like the the financial

collapse

in 2007 and 2008 where you just
absolutely fell off a cliff and none of
the models said that would be possible
or in uh say a psychotic a psychotic break
the most stable guy in the community
just goes totally wild

or good things too uh you know
a religious conversion that turns the
slaver
into the guy who wrote amazing grace
well now

more recently um gerald edelman
emphasizes relatively inaccessible
states a tremendous amount
both in his work on immunology and
in neuroscience the term he uses
is
degeneracy which he borrows i believe
from

analysis numerical analysis to mean
uh something roughly like a function
whose inverse is not a function
of multiple ways of coming up
with an activity which from the outside
looks exactly like another activity but
it's implemented in a completely
different way and then in a
professor fristen's paper uh that was
uh suggested reading a couple of weeks
ago

uh on where he's talking about uh
as uh ascent uh climbing gradient
descent

um he says it seems
as though what's going on inside the
markovian blanket is it's learning about
its environment

but it can't observe the environment how
can it be learning about something that
it can never
observe so i don't know where to go with
that but it all sounds like it kind of
would fit together and make a nice paper
thank you
dave for mentioning that and lots of
great points
stephen and then i'll also go with the
side hook and histories this discussion
yeah i think this one thing this
discussion really talks about is the
the relativity between what the organism
in the markov blanket the generative
model and
the niche and the environment so the
world out there can follow all the rules
of
physics and it could have a lot of side
branches and
it can go and behave in all these sorts
of ways
and which we don't know about some we
can track through the entropy
some we will know because the niche will
hold it i
my cup will still be on the table so
some of the
the kind of the stuff about least action
has a benefit of maintaining a trace
more clearly
in the environment so that there's an
inch i think this is where it's quite
interesting is how much
the the the the way of coming in it from
both directions i'm
starting to see what karl frisco meant

there in some ways is
sometimes it's about the model of the
outside and sometimes it's what's out
there
it's been shaped and humans we're doing
a lot of that
so when we shape something out there
what what is available because they talk
about the adjacent possible
a lot when they're doing work with
people so is it is it that maybe
you know as an animal you know you can
infer
much more diffuse abductive kind of
things if it's within your sensory range
however it needs to be within the
adjacent possible
whereas in the niche you can do things
at distance you know
um you can fire something over the hill
and then you go around next week and you
see where it landed
you know um which is kind of
so anyway i thought that's quite
interesting i think that
um really does speak to this two-way
movement and the fact that some pieces
are more
about you know it changes the name of
the game for emergence for sure
beyond pure physics into this
non-equilibrium
and i'll put a paper because i found a
paper that talked about non-equilibrium
chemistry
when i was looking and it was
interesting it said that there's almost
been no non-equilibrium chemistry

until like they only started doing it in
the 80s and 90s
and it's like then you do it's all this
at equilibrium

type work where you go from one to the
other and then of course all life is a
non-equilibrium so it's kind of um
interesting because you've still got
both going on it's just happening
in the in the niche stephen excellent
point

let me go on just about in reverse order
there's a few excellent things you said
long-range action through niche
modification that's stigmergy the ant
can't do long-range
instant communication but it can
instantly put its abdomen on the ground
and then

three months later another nest could
instantly detect it

so it's at an instant with a behavioral
focus but it's actually long range with
respect to action of that one nest mate
on another

so then there was a discussion by dave
about how fristen was mentioning how
how does the organism learn or how does
the system learn when it can't directly
infer directly you know measure the
states and that's everything we've been
talking about with the observables
the organism is actually observe getting
the observables and using those to do
state estimation on the hidden states of
the world

and then it turns out that the
observables don't have enough

instantaneous information

to reliably identify what you care about

which is the latent causes of the world

so you have this imaginative generative
model

which includes that space of the

adjacency you know the adjacent possible

that's weighted by how likely it is and

a bunch of other stuff

that makes it more like a candle burning
than just

a marble on a hill so it's a little bit

more nuanced because we're talking about

something that's like a bayesian

computation at least as the scientist

uses it

and again that distinction between is

this

quote what's happening realism versus is

this just a instrument that scientists

are using in the world

that's going to come up next week in 15

and so that's like

really interesting all right so the side
hook

i thought if there's a idea with many

sides then maybe it has many

side hooks because like it's clearly by

analogy to the complex number so i like

this idea though of jumping out a

dimension

and that seemed like it was related to

what stephen said with the adjacent

possible

and with policy imagine imagination it's

like if there's a wall in front of you

and then you're only on the number line

you're gonna hit into the wall but then

if you
are logistically in a different
dimension then you can go around it or
you can figure out whatever policy it is
so policy and imagination is like
popping out to a different dimension
where then you can enact
a policy but if your imagination isn't
connected to the manifold of reality
your action selection would it's like a
fantasy it's like if i could fly i could
just leave this building
it's like yeah but the e matrix is not
that way so it can't happen
so it has to be imagination with the
affordances
and then also just one last point on the
statistics
so the markov chain that marco um i
believe
mentioned so like a markov chain with a
memory of one
would mean a statistical process that's
only dependent it only has a memory of
one whatever the previous state was that
is what influences it
and then it's almost like as a statistic
statistical model
so again whether we think of this as a
scientist instrumentally with a model
or the organism actually embodying this
model let's just say that
most things have an influence of one
time scale but then once in a hundred
times it has like a
range of a thousand so there's two
models you could make
you could have a model with a markov

chain of a thousand
so the different action sequences of
like every single possible
last thousand days and that model is
very computationally complicated
but if you've had enough observations
it's unrealistic you'd be able to fit
that like
to a t versus the more simple model
which is just give me the memory of one
so i can have just one bit of memory
and then i'll be surprised when the
other thing happens
and so that is the compromise of the
modeler
of model accuracy minus model complexity
is
do you make this markov blanket like
simple like a heuristic or a bias
and then be surprised by particulars or
do you try to fit
this ridiculous particularized model for
every single case
but then that loses all efficacy because
then you're tied up so specifically with
the very specifics of the model that you
learned
that's like overfitting so i know that's
a ton of points
but those were all like really awesome
comments
and i think we're getting to a lot of
the stops on this little road trip
but let's just head to the last
section but anyone if you have any final
thoughts or questions
that was yeah basically the only other
slide was this previous one with a

little bit of
and then just the usual questions but
i'll just say
thanks for participating and stay in
communication with everyone
and then let's pull back to uh the road
map or to here and then
each of the four of you thanks so much
for these like just amazing
conversations
so each of you if you would like to just
take a you know we'll go one last round
of people
want to so whoever wants to raise their
hand
otherwise we can go to this one i guess
so this slide was kind of starting with
the physics on the bottom
with a focker plank and a flow so that
was what
dave brought up and the way that this
conceptualizes a flow like on a hill you
can have like two orthogonal
curls two orthogonal factors one is kind
of curling around the hill
at the same elevation that's like what
you see on a
camping map like the iso contours and
then the other one is like a ruler
that's
right on the hill like straight up and
down and so
not to go too much into physics but the
ruler
is like the Δg it's like the
gradient descent is the candle burning
or is are we in a flat spot
so that's the ruler is like are you

skiing fast downhill or are you not
skiing fast downhill
and then the iso contour is like is a
steady state
that's like when you're um when you're
doing a chemical reaction
there's always a forward and a backwards
reaction now the K_{eq}
the equilibrium coefficient for the
reaction's completion might be 99.999
percent like a candle but actually the
completion is really high because it's
so exothermic
but stephen from chemical manufacturing
you know that reactions that are barely
favorable
you're going to end up with like 52
percent of the product that you want and
48 that you don't want or something like
that
so those are that's that decomposition
right here with a helmholtz
decomposition
that there's the solenoidal curl and
irrotational or divergence components
so that's kind of a thermo info
connection
so when people are talking about
gradient descent in machine learning
they're talking about like just like
chemicals are allegedly physically
skiing up and down these slopes also in
machine learning
you're doing informational skiing on
these slopes
and then that's where you get these
bayesian statistics and thermodynamic
ideas

the key insight in the the integration
with control theory happens here
basically if you're going to be
maintaining that far from equilibrium
steady state
better than all the other models like
you in the niche
then you need to have a generative model
not just
of state estimation because that's sort
of like
the low hanging not really fruit even if
it is low hanging or not it's just not
really fruit
because action is what the agent needs
to do
to stay far from equilibrium whereas a
machine learning model on our computer
it doesn't need to behave well to
survive because it's like under not
really the kind of selection pressures
that we are
but we need the action element so then
this imperative to be self-evidencing
as an information forager means that you
get all of these cybernetic
characteristics
and that you see systems that act as if
they're doing the optimal
experimentation given their priors
reflected in the generative model in
their niche so
that's where you get this sort of like
forager information
optimal information gain principle
because
that's what's needed to keep the action
selection effective

because they're far from equilibrium
steady state that has to be
subsiding despite these things that's
what gets
subsumed in free energy principle as
discussed in mel's paper and then on the
top
are corollary process theories so it's
like okay
whoa i guess that would have to be true
for any candles i saw
out there how did that energy get so
organized
so it had to be an active candle or it
has to be part of the extended niche of
us
right there's no candles that's like the
watchmaker analogy and then
how does that happen and these are
falsifiable
process theories because theory
falsifiable empirical
unique explanatory explanations and
predictions um
this is where we see the falsification
happening
and so we discussed in this paper like
some of the already kind of i guess
less in favor process theories like the
gibbs filtering or gibbs sampling
which is done on computers but is not
apparently tractable on biological
architectures
in the way that these um more
established or i don't know uh
productive
not falsified yet there's some good
critiques of bayesian brain

in the paper in 15 as well as in um
the 2018 interview in alias with fristen
but um yeah and then what else because
the whole
idea is that like as free energy
principle gets shored up and connected
laterally to a bunch of other things
there's going to be other process
theories
active inference is about some kinds of
systems but who knows there's other ways
we could think about it and
so i don't know what does anyone think
about that other than that we'll head
into our
final section marco
yeah um i guess i can add a hopefully
pleasant perturbation
so one interesting um thing to
to raise here for this slide is
um it's conceivable that active infants
might actually be
more solid in its conceptual framework
or structure
than actually the formalism for example
you could maybe for example
not use kl divergence as a distance
metric
but other distance metrics um because
the kl divergence is i think also in the
self idealization
right in the end you're going to have
very a lot of messy stuff
in in real systems such that even if
you're going to model certain
belief distributions in the action
implementation it's going to be
extremely unlikely

that it perfectly follows this um
a particular conception of distance to
be minimized
um and so i guess i'm just urging
for this expression of what can come out
for me
that that that um
oh okay i personally just like to
encourage people to
do to be playfully exploring different
variants
of the free energy principle because
personally i believe conceptual
uh model uh taking it as a conceptual
model is better
than as purely a formal but that's
personal
yes well thanks for expressing your
thoughts
as an action suggestion stephen and then
sarah
yeah thanks for that description daniel
actually i'm going to go back and be
able to
use that as a really nice reference
point for lots of
information i like this top piece and i
think yeah the
bayesian brain was there as this kind of
integrated idea and then they thought
well let's have this
predictive processing process and um
i think marco was saying like some of
the assumptions then for it to go to
active inference
you know we the math is there to make it
solid but we
you know we can think beyond that

because then they say active inference
is really an

active inference which goes another
level and the math that's not a math
thing that's like okay so if it's n
active inference and then does there
need to be

like you say what else is there i'm
actually quite interested in potentially
been looking at some ideas of some
spatial

layers like we talked to had that thing
about projective consciousness and some
other layers like

what are the what's the layers beyond
inactive that i need to be there for
these processes to be plausible and
who knows that that would be something
that we'll talk about in later date but
i like

this this slide could actually be
something that could be played with in
the future

as a learning tool because i think it's
quite helpful

great thanks so much steven sarah
oh the trail might be kind of cold now
but you super had my attention

um on on yours you know you had your
arrow moving around

and you said you know this is the place
where there's like a connection between
thermal uh energy and free energy and i
was like

where because i mean you know math wise
because i've

that's always just been this super
separate world that's annoying and i

can't really put it together
so i want to find where that is and then
go
find it in the math you know that the
word description in this really
nice slide and then the math and try to
figure that out
great call great call um i agree
it's why one of our projects just to
sort of like talk a little bit about the
lab since everybody here is basically a
participant in it
it's one of the reasons why having the
ontology and the structure of knowledge
for the equations is really important
because we want to be able to say this
equation in this paper
is where the variables in the you know
the
f from this paper in this other math
area actually we're going to call it
this letter
and here's this and that's where there's
a point of contact with this
and then is this one following from that
one is that one following from this one
those are the kinds of things that truly
are
in the frontier of what we can
understand
and then understanding how to bring
directly from these analytical
mathematical phrasings to the control
theoretic in the policy selection
that's a non-trivia it's not a velcro
interface that's a
serious maneuver that involves also
moving through

as pointed out equilibrium and steady
state and far from equilibrium
non-equilibrium all these different
things
in this metaphor or is it not realm of
info thermo so lots of exciting
work to be done and traced good points
and then also yes steven just you
pointed out it's um active inference
imagine if someone said in
2010 well active inference is simply one
time step and it's xyz it's like
it was and then now it's something else
we've talked about
and activism we've talked about
affective inference
now is the variable in affective
inference
equivalent to answering philosophical
questions about affect no
but neither was the variable affect in
any other modeling framework
which again makes us pull back and just
look at that instrumentalism versus
realism debate
so just the realism would be this is
actually telling us
how the world is arranged and what it's
doing versus just
this is a framework that is being
just like mel's paper this is one of the
main points is
with respect to science it's in the
realm of science
but what is it in the realm of science
well
it's not one of these falsifiable
theories up here

it's something that's more like natural selection
at that type of tier but it's like a theory generator and and that's why if we look at the road map there's um 14 sections on kinds of models because this is like the system that this is being slotted into
mel could have or anyone potentially could go the other and say no it's not part of science it's part it's okay so you can debate about where it is with respect to the demarcation problem but since this is the paper we're in the journal club on we're saying given that that's the view being defended the literature has to be integrated in is not theology it's not poetry
it's actually scientific modeling philosophy of science literature so then that will help us connect it to like
okay but what does the variable in a linear regression mean
is that realist versus instrumentalist interpretation nobody would say it's realist it's not there's not like a line in nature it's making one you know group of plants a different shade than another group of plants so that's sort of the process that this paper is engaged in
is interfacing better so that we can understand what kind of thing the fep is and uh there's a few

uh exhortations like this is it true
what is it true of
how do we empirically know that it's
true and mel
argues that that's a category mistake so
it's like saying
any other category mistake now think
about these questions in relationship to
the fep
being like that so again great
paper we'll just go to the closing
thoughts
now but just thanks everyone for
being a part of 14.2 and all the other
sections because this was really like
two great discussions
i'm looking forward to that people can
raise their hand if they want to make
final comments but
i'm really looking forward to the
discussion
with the authors um hippolyto and van
s on 15 for both weeks and that will be
interesting and
we'll confront this realism
instrumentalism discussion
and then also um the zero videos will be
a dialogue so hopefully more and more
people will
get involved with making the summaries
of the videos
and introducing them from different
perspectives different lengths
and each paper is unique instead of
authors and set of people who are
approaching it so i just
it's been really awesome to see people
engaging with some of these

truly long-standing and fundamental
questions
in a new light and then also
just co-creating with
first just participation in live streams
maybe even just by watching but then
anybody who's watching
live earned replay is always welcome to
join or find out what we're doing in the
lab
so it's just been a really good you know
we grew a lot during 14
and so is good times so any other final
comments that
the panel wants to make yeah i'm gonna
i'm gonna grow some neurons now this was
awesome peace
marco go for it yeah i just wanted to
say
um ever since i first joined i can't
find
the survey you keep mentioning i have no
idea where the survey is so maybe
get to tell us and the viewers
where to provide feedback sure well the
survey
for live participants is in the event
calendar
so it's in the calendar invite and so
yeah for those who are listening to the
very bitter end the drags of the active
stream
basically the single source of truth is
the calendar invite
it has the slides that are accommodable
the paper
the video url and the feedback form and
then the feedback form

you know it'd be awesome if everybody
fills it out but then the feedback form
for everyone else is just
email comment threads youtube comments
just
leave feedback publicly or privately
we'll always respect
both public and private feedback and let
you know how it influenced our
trajectory so but for live participants
we have a special structured form
that we can always modify and evolve as
well so
that's a little insider baseball but
always happy to talk about that because
that's
what it is we do is it not steven
and sarah on the discord and all the
decision making
maybe it is maybe decent but it's been
fun
so no it's great a very good explanation
and uh
i enjoy it today right like so i've got
to grow i think i'll grow a spare brain
and then i'll fill it up and put it back
in my head
just force it off source it put it in
the papes
all right good to uh talk to you all i'm
gonna terminate the live stream